

Vikas Chand

Curriculum Vitae

PhD in Physics

Thesis : Spectro-polarimetry of Gamma Ray Bursts

Supervisor : Prof. A. R. Rao

Department of Astronomy and Astrophysics at Tata Institute of Fundamental Research-Mumbai

Postdoctoral Researcher: ARCO Israel (Jan, 2022 - present)

Supervisors: Prof. Jonathan Granot, Dr. Paz Beniamini

Postdoctoral Research Fellow: Nanjing University, China (Dec, 2019 - Nov 2021)

Supervisor: Prof. Binbin Zhang

Courses passed at TIFR Graduate School

- Computational Physics, Quantum Mechanics-II, Gravitation and Cosmology, Statistical Mechanics, Advanced Quantum Mechanics, Astronomy and Astrophysics-I&II, Electrodynamics-II, Experimental Methods, and Fluid Mechanics-I

List of Publications

In Preparation

- Orphan GRB Afterglows – Realistic Predictions and Survey Strategies, Vikas Chand, Jonathan Granot and Paz Beniamini (in prep.)

Published papers

- Prompt Emission Polarimetry of Gamma Ray Bursts with AstroSat CZT-Imager, Tanmoy Chattopadhyay, .. , **Vikas Chand** et al. ApJ, Vol 884, 2, 123, 20 (2019)
- Peculiar Prompt Emission and Afterglow in H.E.S.S.-detected GRB 190829A, **Vikas Chand** et al. ApJ 898, 42, 13pp. (2020)
- GRB 200415A: A Short Gamma-Ray Burst from a Magnetar Giant Flare?, Jun Yang, **Vikas Chand** et al. ApJ, Vol 899, 106 11pp. (2020)
- AstroSat CZT Imager Observations of GRB 151006A: Timing, Spectroscopy, and Polarisation Study, A. R. Rao, **Vikas Chand** et al. ApJ, Vol 833, 1, 86, 10 pp. (2016)
- AstroSat-CZTI Detection of Variable Prompt Emission Polarization in GRB 171010A, **Vikas Chand** et al. ApJ, Vol 874, 1, 70, 23 pp. (2019)
- Violation of Synchrotron Line of Death by the Highly Polarized GRB 160802A, **Vikas Chand** et al. ApJ, Vol 862, 2, 154, 11 pp. (2018)

7. MAGICal GRB 190114C: Implications of Cutoff in the Spectrum at sub-GeV Energies, **Vikas Chand** et al., ApJ, Vol 903, 9, 12pp. (2020)
8. Synchrotron and synchrotron self-Compton emission components in GRBs detected at very high energies, JC Joshi, **Vikas Chand**, S Razzaque, World Scientific Publishing Co. Pte. Ltd., 2023. ISBN #9789811269776, pp. 3009-301
9. Surprise in Simplicity: an Unusual Spectral Evolution of a Single Pulse GRB 151006A, Basak, R.; Iyyani, S.; **Chand, V.** et al. MNRAS, Vol 472, 1, p. 891-903, (2017)
10. Magnetar Giant Flare Originating from GRB 200415A: Transient GeV Emission, Time-resolved $E_p - L_{iso}$ Correlation and Implications, **Vikas Chand** et al. 2021, RAA 2021 Vol. 21 No. 9, 236 (7pp)

Presentations

- "Realistic Predictions and Survey Strategies for Orphan GRB Afterglows Using a Semi-Analytic Model Calibrated by Numerical Simulations", GRB Forum 2024, Greece
- Poster: "Orphan GRB Afterglows – Realistic Predictions and Survey Strategies", GRB50: The Past, Present, and Future of Gamma-Ray Bursts, 2023, US
- "Implications of Cutoff observed in Fermi-LAT GRBs spectrum", Tenth International Fermi Symposium, 2022, South Africa
- Poster presentation: "MAGICal GRB 190114C: cutoff in the spectrum at sub-GeV energies", Yamada Conference LXXI: Gamma-ray Bursts in the Gravitational Wave Era 2019, Japan
- "Probing GRBs with prompt emission spectro-polarimetry", 2019 Nanjing GRB Conference, China
- "AstroSat/CZTI - Spectro-polarimetric study of GRB 160802A and GRB 171010A", VHEPU 2018, Vietnam
- Poster presentation: "Spectral, Timing and Polarization properties of GRB 160509A & GRB 160802A: a tale of two GRBs with polarization measurements by AstroSat/CZTI", Playa del Carmen, Mexico
- Poster display: "Spectral and timing analysis of first GRB observed with Astrosat", Cargese School 2016, France
- "Deciphering the GRBs with spectral, timing and polarization measurements" at Workshop on Gamma-ray Bursts : Prompt to Afterglow 2017, TIFR - NCRA Pune
- "Can GRBs be standard candles?" at Wide Band Spectral and Timing Studies of Cosmic X-ray Sources, 2016, TIFR, Mumbai
- "GRBs with *Swift*-BAT and *Fermi*-GBM" at CZTI mini workshop IUCAA, 2015

Schools and Workshops

- Radio Astronomy School 2015, NCRA TIFR
- Cargese School 2016, France

- AstroSat-CZTI mini workshop IUCAA 2015, India
- GROWTH Winter School 2018, IITB, Mumbai, India
- 1st Thai-CTA Workshop on Astroparticle Physics 2019, Thailand

Awards and Fellowships

- 2024 NJU International Fellowship for Senior Scholars, School of Astronomy and Space Science, Nanjing University, China
- 2018 Sarojini Damodaran Fellowship, Tata Institute of Fundamental Research (TIFR), Mumbai, India

Education

- Jan 2013 – 2019 **Research Scholar**, *Tata Institute of Fundamental Research*, Mumbai, (with two years of coursework).
- July, 2012 – Jan, 2013 **Junior Research Fellow**, *Physcial Research Laboratory*, Ahmedabad.
- 2010–2012 **Masters of Science**, *Panjab University*, Chandigarh, *percentage – 69.35 %*.
(June, 2012) First Class Honours
Specialized in Astrophysics
- 2007–2009 **Bachelor of Science**, *Post Graduate Govt. College 11*, Chandigarh, *percentage – 79.9 %*.
- 2006 **Senior secondary**, *Crescent Public Sen. Sec. School Palampur*, Himachal Pradesh, *percentage – 75.6 %*.
Eighty sixth(y) position in merit

Competitive exams

- GATE 2012, All India Rank (AIR)-179
- JEST 2012, AIR-358
- CSIR NET June 2012, AIR-23
- BARC entrance exam for Ph.D.

Other Exams

- Combined Defence Services Exam (CDSE)
- Fast Track Scheme of Indian Airforce

Hobbies

Mountaineering: Completed Basic and Advanced Courses (A-grade) at ABVIMAS, Manali.

Paragliding: Courses P1, P2, P3 and thermaling training.